



Tag 15

Freitag 10.07.2026

Von KI erstellt, von Menschen kontrolliert.

Hallo Mannheim! Hallo Heidelberg!

GitHub Repo <https://github.com/www42/a5>

AZ-104

Learning Paths:

- Manage identities and governance in Azure
- Prerequisites for Azure administrators – CloudShell and templates
- Configure and manage virtual networks for Azure administrators
- Implement and manage storage in Azure
- Deploy and manage Azure compute resources
- Monitor and back up Azure resources



TCP/IP Vint Cerf 1964



AZ-104

 Deploy and manage
Azure compute resources

Compare Containers to Virtual Machines

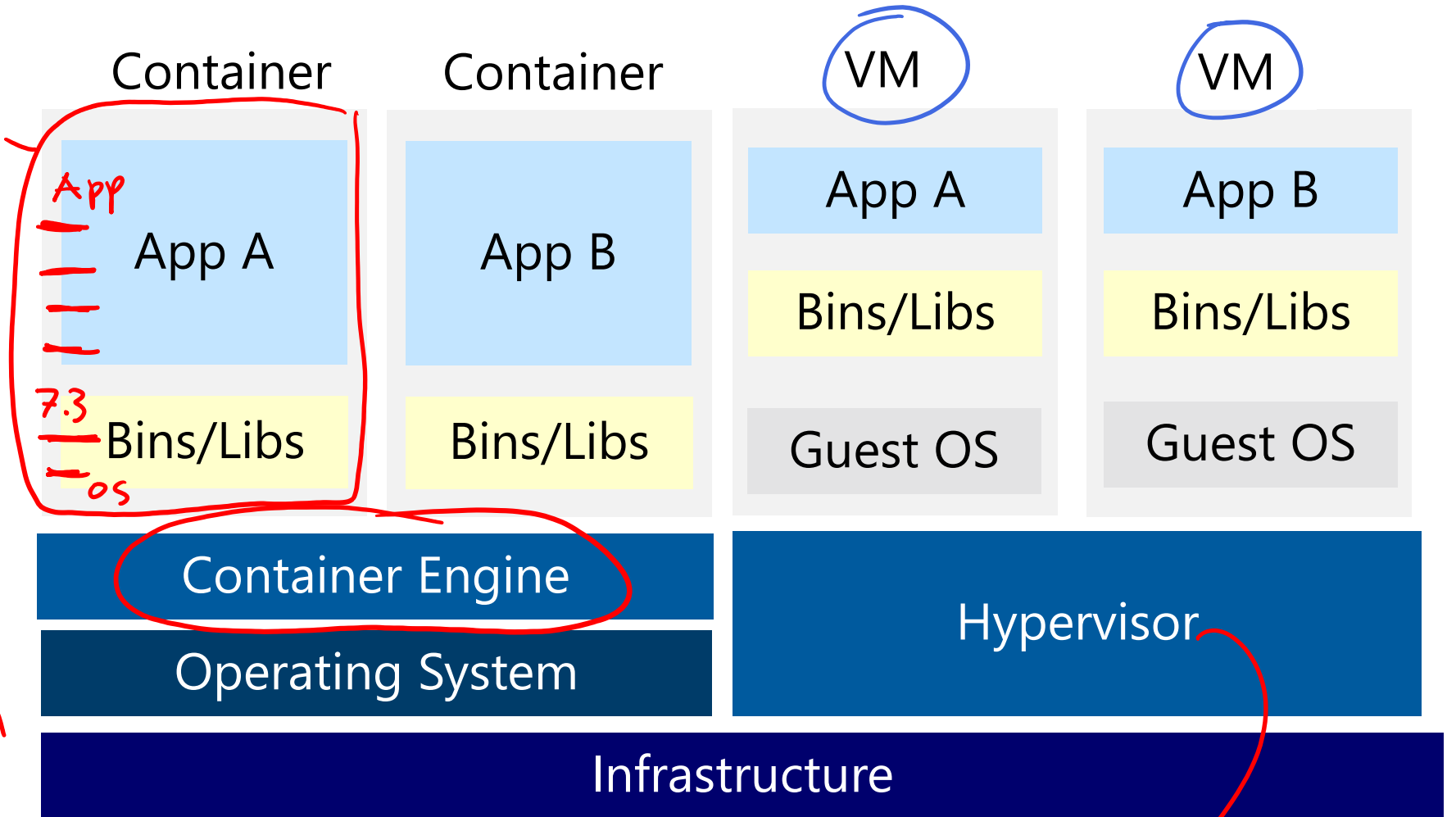
Light weighted CaaS ← Heavy IaaS

Docker Desktop
NAS
AWS
Azure ACI

- Isolation
- Operating System
- Deployment
- Persistent storage
- Fault tolerance

Docker Image

Containerd



Hyper-V
VMware
KVM Linux ...
WSL
Proxmox

Heute

- einzelne Container (Azure: ACI ACR Instance Registry)
- Lokal: docker version Lab
- Kubernetes (griechisch: Steuermann)
= Orchestrierung von Containern (Pod)

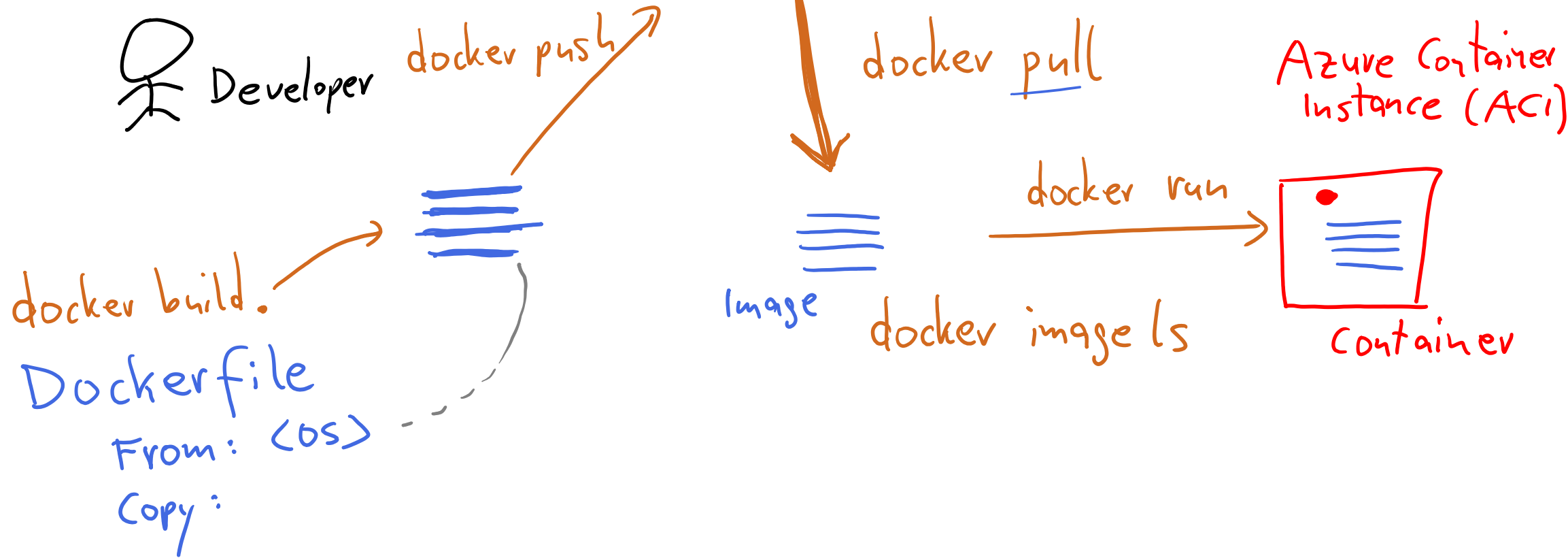
Lab in Azure for Students

Mannheim
Luisenpark
Beo

Quickstart

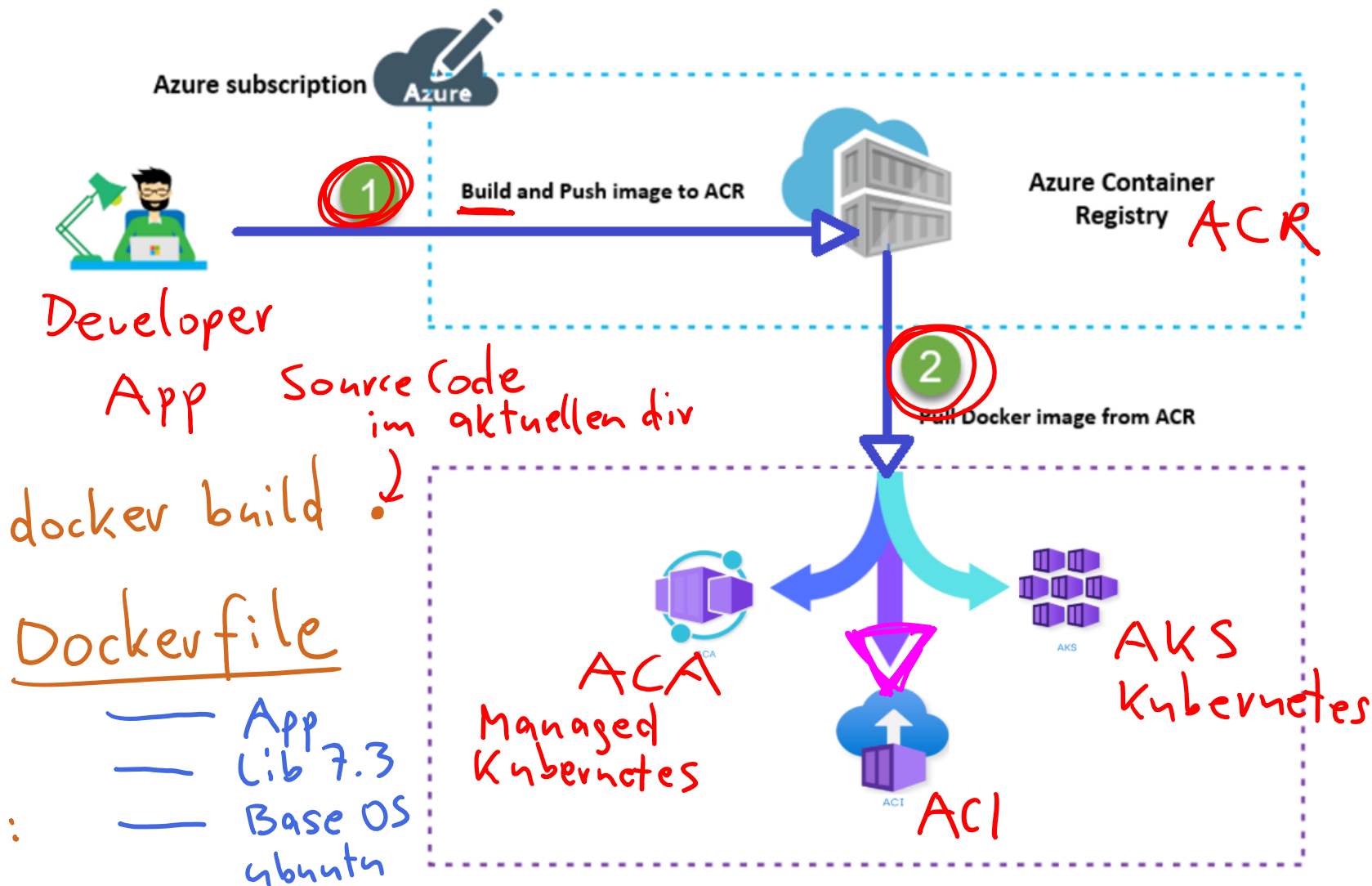
Registry

Docker Hub
Azure Container Registry (ACR)



Understand Container Images

A container image is a lightweight, standalone, executable package of software that encapsulates everything needed to run an application.



Lab (Azure for Students)

GitHub A5

Image bauen

Entweder: Lokal
mit Docker Desktop

`docker build`

Dockerfile

Oder: in Azure
mit ACR
(geht nicht ?)

`az acr build`

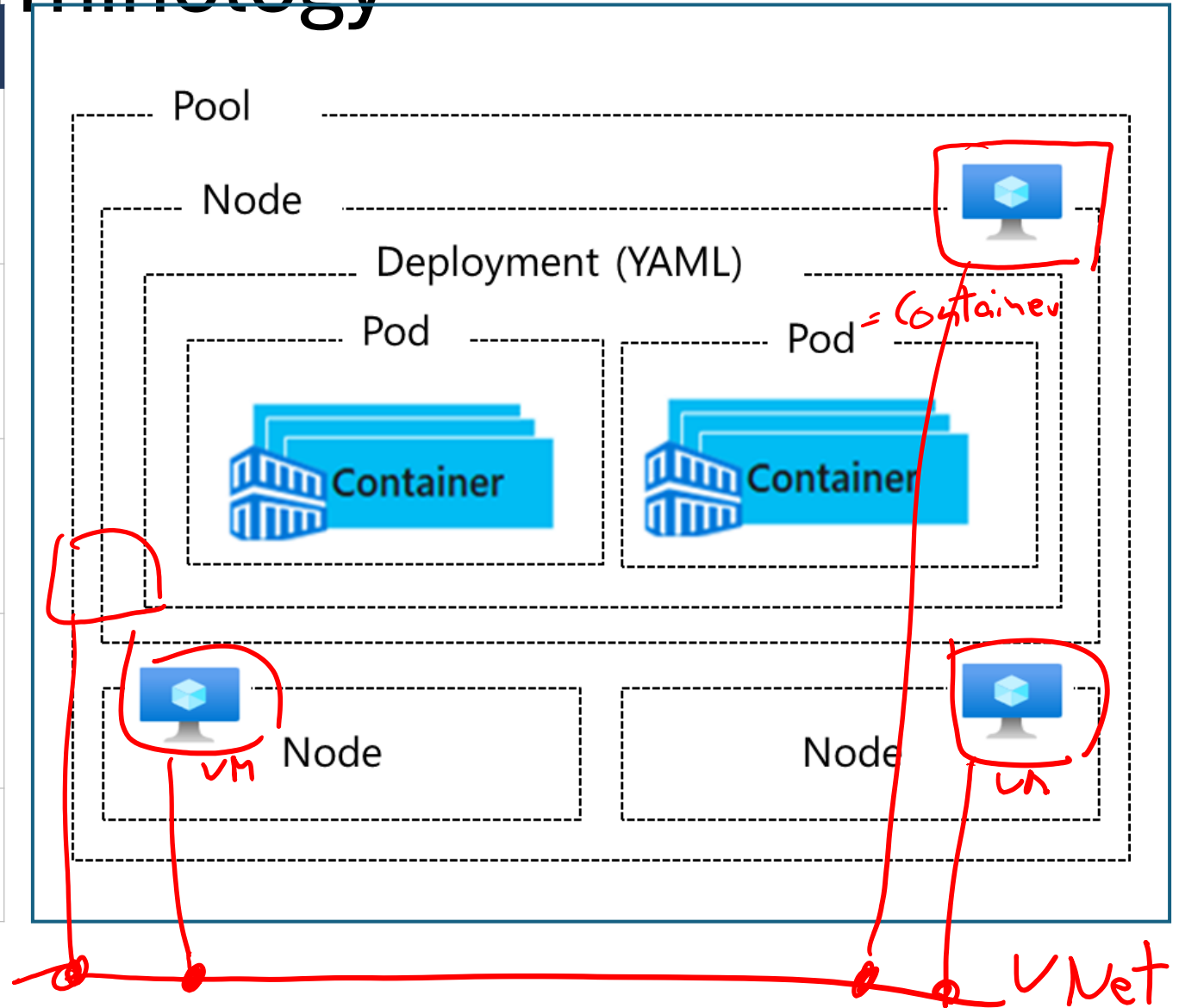
Kubernetes (Google: Borg, Brendan Burns + 2
7/9 Kubernetes

Understand AKS Terminology

Term	Description
Pools	Groups of nodes with identical configurations
Nodes	Individual VMs running containerized applications
Pods	Single instance of an application. A pod can contain multiple containers
Deployment	One or more identical pods managed by Kubernetes
Manifest	YAML file describing a deployment

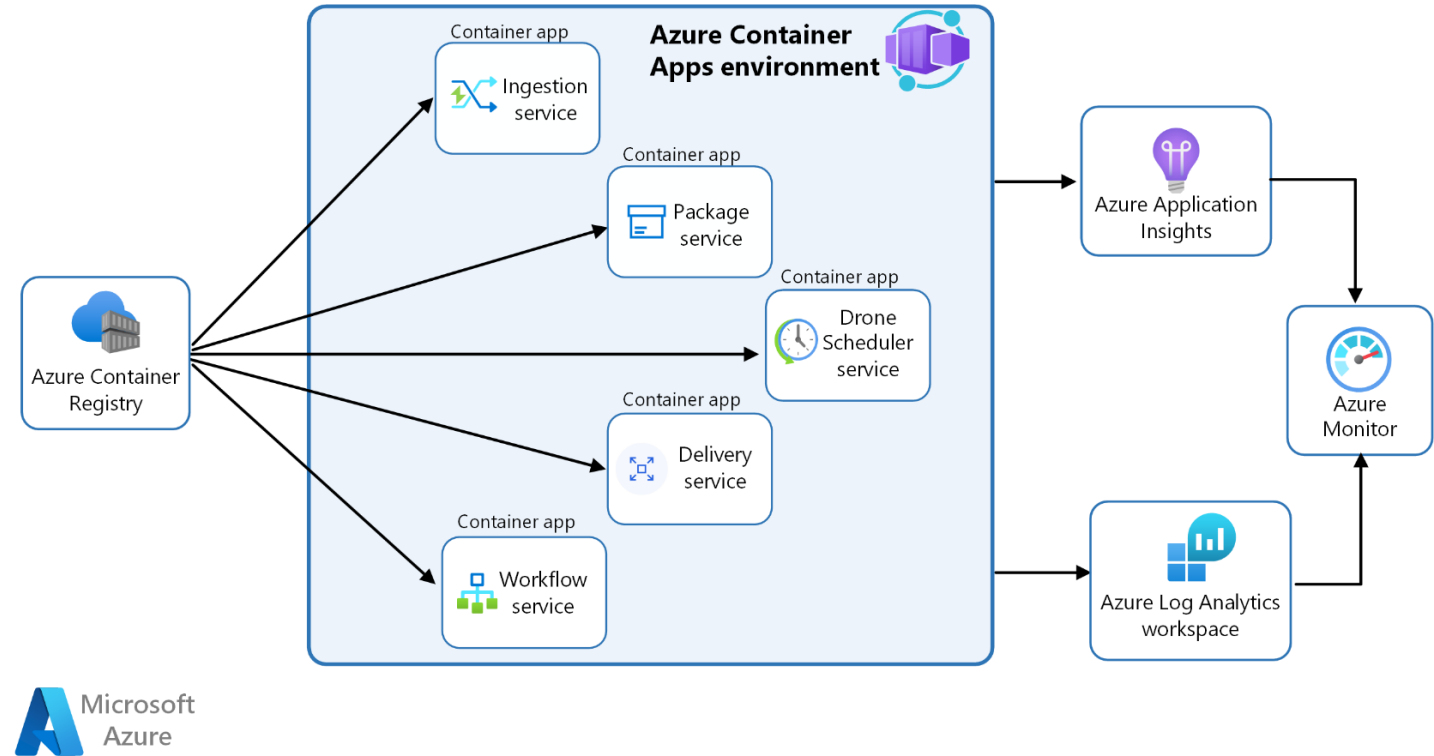
API

Master



Manage Containers with Azure Container Apps

- Alternative to Azure Kubernetes Service – manages container orchestration
- The Container App environment creates a secure boundary around the apps and jobs
- The Container App runtime manages the environment (OS upgrades, scaling, versioning, and failover)



Compare container management solutions

	Azure Container Apps	Azure Kubernetes Service
Overview	Simplifies the deployment and management of microservices-based applications by abstracting away the underlying infrastructure.	Simplifies deploying a managed Kubernetes cluster in Azure by offloading the operational overhead to Azure.
Deployment	PaaS experience.	Offers more control and customization.
Management	Fully managed by Azure.	Partially managed by Azure (control plane).
Scalability	HTTP-based autoscaling and event-driven scaling.	Horizontal pod autoscaling and cluster autoscaling.
Use Cases	Rapid scaling and simplified management.	Complex, long-running applications that require full Kubernetes features.
Integration	Azure Logic Apps, Functions, and Event Grid for event-driven architecture.	Azure Policy for Kubernetes, Azure Monitor for containers, and Azure Defender for Kubernetes for comprehensive security and governance.

Labs



TO THE LAB ZONE →

NO THROUGH ROUTE

Berlin
Europa
Dublin

SPEEDY ROUTE

4²

lounge

Microsoft
tech-ed
2010

lounge

software diagnostics

E94

SD Studio

for Team Foundation Server
Visualize your code in
a software map.

SD Developer Edit

for C and C++ for Visual
Trace dynamically your code
Visualize the control flow.

software
diagnostics

DataCore
SOFTWARE

End of presentation



